

The TOR signaling pathway regulates vegetative development, aflatoxin biosynthesis and pathogenicity in *Aspergillus flavus*

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Abstract

The target of the rapamycin (TOR) signaling pathway is highly conserved and important in eukaryotes. It is involved in the regulation of various biological processes. However, systematic studies on this pathway in the genus *Aspergillus* have not been reported. Here, we identified and characterized nine genes encoding components of the TOR pathway in *A. flavus*, and investigated their biological, genetic and biochemical functions. The FK506-binding protein FKBP3 and its lysine succinylation are important for aflatoxin production and rapamycin resistance. The Tor kinase plays a central role in the global regulation of growth, spore production, aflatoxin biosynthesis and rapamycin stress. As a major downstream effector molecule of Tor kinase, the Sch9 kinase might regulate the calcium and osmotic stress, AFB₁ synthesis of *A. flavus* by its S_TKc, S_TK_X domains and ATP binding site at K340. We also showed that Sch9 kinase might mediate crosstalk between the TOR and the HOG signaling pathways. TapA and TipA, the other downstream components of Tor kinase, play important roles in regulating mycelial growth and sclerotia formation in *A. flavus*. The member of the TapA-phosphatase complexes Sit4 and Ppg1 are important for hyphal development, sexual reproduction, sclerotia formation, AFB₁ biosynthesis, activation of the CWI and TOR signaling pathways in *A. flavus*. In addition, the another phosphatase complex Nem1/Spo7 play critical role in vegetative growth, conidiation, aflatoxin and LD biogenesis. This study provide new insights into constructing the regulatory network of the TOR signaling pathway and revealing the molecular mechanism of the pathogenicity in *A. flavus*.

eLife assessment

This **useful** manuscript describes the TOR signaling pathway in the human and plant pathogen *Aspergillus flavus*. While the authors provide a large amount of descriptive and often confirmative data, the evidence for the new claims made here for the TOR pathway in this species is **incomplete**.

Introduction

A. flavus is one of the most important phytopathogenic fungi species, which frequently contaminates various plants, causing economic losses in agricultural commodities and posing serious health concerns in developing countries [1]. *A. flavus* is also an opportunistic human pathogenic fungus that can cause aspergillosis in immune-compromised patients [2]. Aflatoxins (AFs), the toxic and carcinogenic secondary metabolites produced by *A. flavus*, are the most potent carcinogen present in nature, causing liver cancer in humans and animals [3]. Although most of the enzymatic reactions involved in AF biosynthesis have been elucidated and the genes encoding these enzymes have been isolated [4], the signaling pathway network and specific regulatory processes involved in AF biosynthesis are still unclear. Therefore, exploring the regulatory mechanism of the AF biosynthesis signaling pathway would provide new insights for preventing *A. flavus* and aflatoxins contamination.

In eukaryotes, the target of the rapamycin (TOR) signaling pathway is highly conserved and plays essential roles in several important biological processes, such as ribosome biosynthesis, cell growth and autophagy [5]. Several studies have shown that the TOR signaling pathway interacts with other signaling pathways (MAPK, CWD), and they crosstalk to form a complex metabolic network to regulate cellular processes [6–9]. The serine/threonine-protein kinase Tor acts as a crucial protein element in the TOR signaling pathway, which interacts with other proteins to form multiprotein complexes. There are two TOR complexes named TORC1 and TORC2 in *Saccharomyces cerevisiae* [10]. TORC1-mediated signaling controls cell growth by regulating many growth-related processes and is sensitive to rapamycin, while TORC2-mediated signaling is mainly responsible for cytoskeletal remodeling and is insensitive to rapamycin [11]. However, most organisms, including plants, animals and filamentous fungi, have a single Tor kinase [12–13].

In the fungal kingdom, most studies on the TOR signaling pathway have been conducted in budding yeast *S. cerevisiae* [14–15], and fission yeast *Schizosaccharomyces pombe* [6,16–17]. In *S. cerevisiae*, rapamycin forms a complex with the peptidylprolyl cis/trans isomerase FKB12, which binds and inhibits Tor kinase. The Tap42 phosphatase complex is the main target of Tor kinase, and the Tap42 protein interacts with the Tip41 protein to jointly regulate phosphatases such as Pp2A and Sit4. Phosphatase Sit4 can dephosphorylate transcription factor GLN3, regulating the nitrogen source metabolism pathway [18–19]. In addition, the other immediate effector of TORC1 is AGC kinase Sch9, which participates in other cellular functions including ribosome biosynthesis [20]. In filamentous fungi, the TOR signaling pathway has been reported in *Fusarium graminearum* [21–22], *Magnaporthe oryzae* [23], *Phanerochaete chrysosporium* [24], *Podospora anserina* [25], *A. nidulans* [26], *A. fumigatus* [51–52], *F. fujikuroi* [27], *Botrytis cinerea* [28], *F. oxysporum* [29] and *Mucor circinelloides* [30]. In a word, filamentous fungi TOR signaling branch mediates spatial and temporal control of cell growth by activating anabolic processes such as ribosome biosynthesis, lipid droplet biogenesis, carbon and nitrogen metabolism, protein, lipid, and nucleotide synthesis, and repressing catabolic processes such as autophagy and apoptosis.

In yeast and filamentous fungi, the TOR signaling pathway modulates cell growth and vegetative differentiation by sensing changes in external nutrients, energy and stress [31]. While the underlying mechanisms of multiple crosstalks between TOR and the other signaling pathway in *A. flavus* pathogens are unclear. Therefore, we intended to identify the gene of TOR signaling pathway and demonstrate their roles and contributions in the regulation of vegetative development and aflatoxin biosynthesis in *A. flavus*. In this study, the results indicated that the TOR signaling pathway plays crucial roles in various cellular processes including vegetative development,

aflatoxin biosynthesis and pathogenicity in *A. flavus*. Our results also indicated a highly complicated cross-regulatory relationship between TOR and other signaling pathways (MAPK, CWI) that maintain cellular growth and survival in response to environmental stress in *A. flavus*.

Results

Rapamycin inhibits *A. flavus* growth, sporulation, sclerotia and aflatoxin production

Rapamycin, a secondary metabolite produced by *Streptomyces hygroscopicus*, is useful in the inhibition of certain filamentous fungi. In *F. graminearum*, rapamycin exhibited a strong inhibitory effect on growth and asexual reproduction [21], which also induces lipid droplet (LD) accumulation in *F. graminearum* and other filamentous fungi [22]. To investigate the effect of rapamycin on vegetative development, secondary metabolism and LD accumulation of *A. flavus*, we assayed the sensitivity of the WT strain to rapamycin. We found that the *A. flavus* WT strain was very sensitive to rapamycin, and the mycelial growth, the germination of dormant spores and conidia formation of the WT strain were significantly inhibited on PDA solid plates supplemented with 100 ng/mL rapamycin (Fig. 1A). The radial growth and the spore production of *A. flavus* WT strain were gradually decreased with the increasing concentration of rapamycin (Fig. 1D, 1F). In addition, the rapamycin also showed intense inhibition on sclerotia and AFB₁ biosynthesis. As shown in Fig. 1B and 1C, compared to untreated strains, *A. flavus* was unable to synthesize aflatoxin and form sclerotia in PDA added with 100 ng/mL rapamycin. Furthermore, we observed that *A. flavus* displayed a dramatic reduction in the rate of spore germination and sparser conidiophores with rapamycin treatment (Fig. S2A, S2B). Calcofluor white is a chemifluorescent blue dye that is nonspecifically used to bind to the chitin of fungi. Fluorescence microscopy showed that the blue fluorescence on the cell wall is significantly diminished when treated with rapamycin (Fig. S2C). The intracellular lipid droplets were stained by BODIPY in the hyphae, and we found that the hyphae treated with rapamycin contained many more LDs (Fig. 9A). These findings suggested that rapamycin-treated hyphae led to the accumulation of LD biogenesis and reduction of chitin content. To explore the potential regulatory function of the TOR pathway to affect the growth, sporulation, sclerotia and aflatoxin biosynthesis of *A. flavus*, qRT-PCR was used to determine the expression levels of genes related to conidia, sclerotia and aflatoxin biosynthesis. The transcription levels of sporulation and sclerotia genes were both significantly downregulated in rapamycin treatment strains, while the genes related to aflatoxin synthesis were almost not expressed (Fig. 1G–1I), which indicated that the TOR pathway is involved in growth, sporulation, sclerotia and aflatoxin biosynthesis in *A. flavus* by regulating these genes. All the above results showed that rapamycin affects the growth, sporulation, sclerotia and aflatoxin synthesis of *A. flavus*, indicating that the TOR pathway plays important roles in vegetative growth and secondary metabolism in *A. flavus*.

Effect of FKBP3 and its lysine succinylation on AFB₁ biosynthesis and rapamycin resistance in *A. flavus*

FK506-binding proteins (FKBPs) are highly conserved immunophilin family and its members involved in various cellular functions including protein folding, immunosuppression, signaling transduction and transcription [33]. FKBPs act as molecular switches, which elicit function through binding and altering the conformation of target proteins. In *S. cerevisiae*, FKBP12 and rapamycin form a complex that negatively regulates the Tor kinase activity [33]. According to the protein sequence of peptidylprolyl isomerase fprA from *A. nidulans*. Four genes encoding putative proteins with FK506 binding domain, named FKBP1 (AFLA_126880), FKBP2 (AFLA_087480), FKBP3 (AFLA_010910) and FKBP4 (AFLA_128200) were identified from the *A. flavus* genome. These FKBPs contain a single FKBP-type peptidylprolyl cis-trans isomerase conserved domain. Disruption

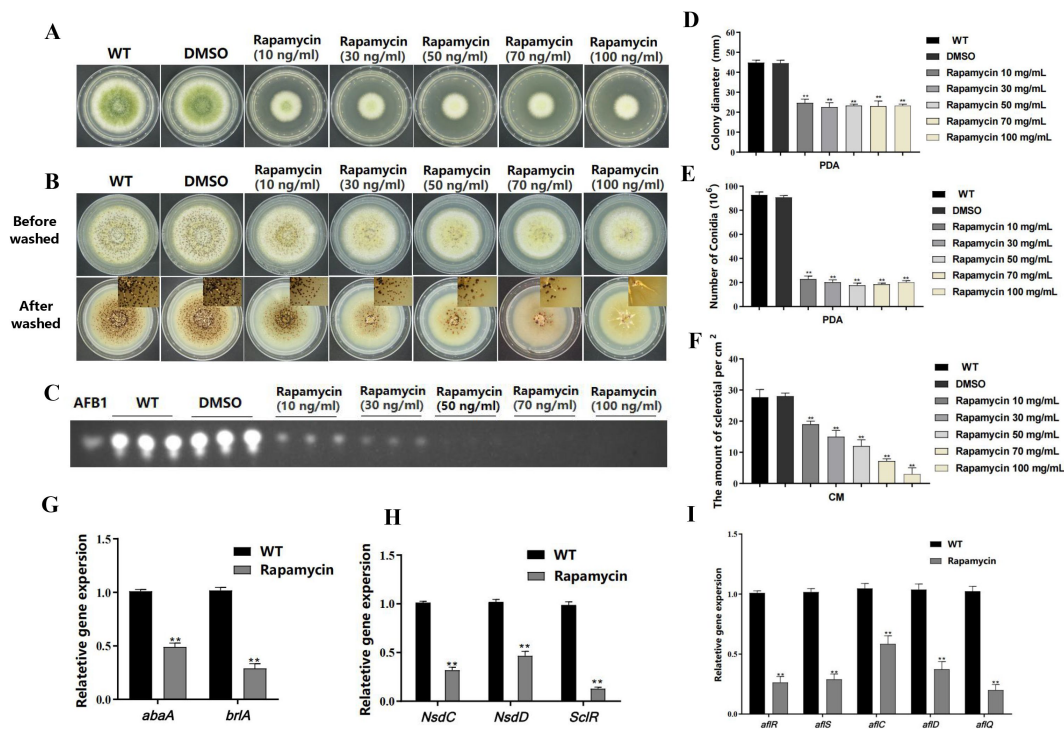


Fig. 1

Impacts of rapamycin on growth, sporulation, sclerotia and aflatoxin production of *Aspergillus flavus*.

- (A) The *A. flavus* wild-type (WT) strain was incubated on PDA amended with different concentrations of rapamycin at 37 °C for 5 days.
- (B) Colony phenotype of WT strain grown on CM medium amended with different concentrations of rapamycin at 37 °C for 7 days.
- (C) TLC assay of AFB₁ production by WT strain cultured in YES liquid media at 29°C for 6 days.
- (D) Statistical analysis of the diameter by the WT strain treated with different concentrations of rapamycin described in (A).
- (E) Conidial quantification of the WT strain treated with different concentrations of rapamycinas as mentioned in (A).
- (F) Quantitative analysis of sclerotium formation by the WT strain treated with different concentrations of rapamycin described in (C).
- (G) Relative transcript levels of conidia-related genes in WT strain treated with 100 ng/mL rapamycin cultured for 48 h.
- (H) Relative transcript levels of sclerotia-related genes in WT strain treated with 100 ng/mL rapamycin cultured for 48 h.
- (I) Rative transcript levels of aflatoxin biosynthesis regulatory and structural genes in WT strain treated with 100 ng/mL rapamycin cultured for 48 h.

strains for each FKBP gene were generated by homologous recombination using the fusion PCR approach. All FKBP disruption strains were confirmed by PCR and sequencing analysis. Morphological analyses showed that mycelial growth and conidia formation of all the FKBP mutants were similar to that of the WT strain on the PDA medium, suggesting that FKBP had no significant effects on the vegetative development (Fig. 2A). However, the number of sclerotia was significantly reduced in all FKBP mutants compared to that of the WT strain (Fig. S3), indicating that FKBP was essential for sclerotia formation. We also assessed AFB₁ production via thin layer chromatography (TLC), and the results showed that the deletion of FKBP3 led to an obvious decrease in AFB₁ production compared to that of the WT strain (Fig. 2B). Taken together, these data indicated that FKBP is involved in sclerotia and aflatoxin biosynthesis in *A. flavus*.

We found that the *A. flavus* WT strain was very sensitive to rapamycin. To determine which FKBP is involved in rapamycin sensitivity, all mutants were cultured in the PDA solid plates supplemented with 100 ng/mL rapamycin. The FKBP3 disrupted strain showed high resistance to rapamycin, but the FKBP1, FKBP2 and FKBP4 deletion mutants showed no change in resistance to rapamycin (Fig. 2A). These results demonstrated that FKBP3, rather than FKBP1, FKBP2 and FKBP4, is a major target for rapamycin in *A. flavus*. For FK506-sensitivity analysis, we found that mycelial growth and conidia formation of *A. flavus* were severely inhibited on PDA added with 10 ng/mL FK506, and the deletion of FKBP3 resulted in increased resistance of *A. flavus* to FK506 compared to WT strain (Fig. 2A), indicating that the FKBP3 was mainly involved in the regulation of FK506 resistance.

In a previous study, we accomplished a global succinylome analysis of *A. flavus* [60]. Based on our succinylome result, we identified six reliable succinylation sites (K5, K19, K40, K42, K55, K65) on FKBP3. Further evolutionary conservation analysis revealed that lysine at 19 of FKBP3 was highly conserved in the *A. flavus* orthologs, suggesting that this residue of FKBP3 may be important for an evolutionarily conserved function, which may play a crucial role in regulating aflatoxin biosynthesis in *A. flavus*. To these succinylation sites, the site-directed mutations were constructed according to the homologous recombination strategy. Compared with the *A. flavus* WT strain, all point mutants exhibited similar phenotype in vegetative growth and conidiation. Interestingly, we observed that K19A displayed increased resistance to rapamycin, and AFB₁ production was significantly reduced when compared to the WT strain (Fig. 3E, 3G), suggesting an essential role of K19 succinylation of FKBP3 in rapamycin resistance and aflatoxin biosynthesis. In addition, other point mutants, including K40A, K42A and K55A, produced less AFB₁ compared to the WT strain (Fig. 3G), illustrating that the succinylation of FKBP3 on others site are involved in AF biosynthesis in *A. flavus*. However, the K19 site likely plays a more prominent role than others sites in FKBP3 succinylation.

Tor kinase plays important roles in *A. flavus*

In *S. cerevisiae*, the FKBP12 and rapamycin forms a gain-of-function complex interacts specifically with the TOR proteins. The Tor kinases can bind to various proteins to form complexes such as TORC1 and TORC2. TORC1 can sense signals such as nutrients, growth factors, and stress in the environment, and participate in the regulation of signaling pathways such as gene transcription, protein translation, ribosome synthesis and autophagy [48]. To identify the ortholog of the *S. cerevisiae* TOR in *A. flavus*, the *S. cerevisiae* TOR protein was used as search queries by BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) in the *A. flavus* genome database. The *A. flavus* genome has a single Tor kinase ortholog (AFLA_044350) encoding a protein with 48.91% similarity to *S. cerevisiae* TOR2. To determine the function of Tor kinase in *A. flavus*, we targeted *Tor* for gene deletion. While we failed to obtain a null mutant, we constructed a single copy deletion strain $\Delta Tor1^+/Tor2^-$ (containing one copy of the *Tor* gene) constructed by homologous recombination. Compared with the WT strain, the $\Delta Tor1^+/Tor2^-$ strain grew significantly slower on PDA (Fig. 3C), and produced fewer conidia and sclerotia than the WT strain (Fig. 3B, S4). The results

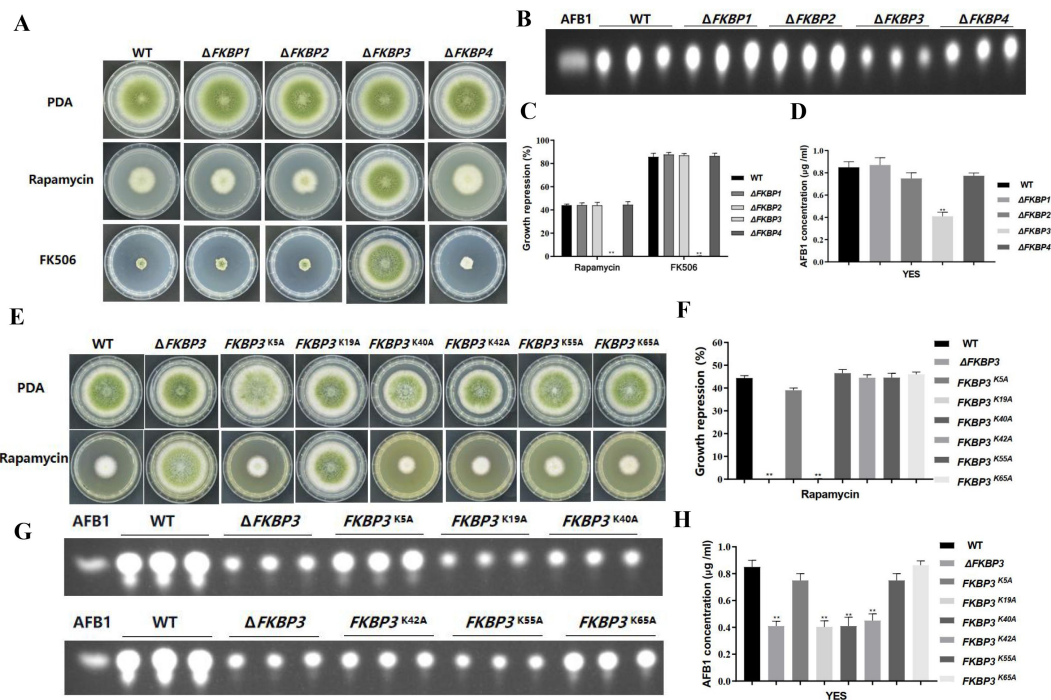


Fig. 2

Disruption of FKBP3 significantly increases the resistance of *A. flavus* to rapamycin and FK506.

(A) Phenotype of *A. flavus* WT and all mutant strains ($\Delta FKBP1$, $\Delta FKBP2$, $\Delta FKBP3$ and $\Delta FKBP4$) grown on PDA amended with 100 ng/mL rapamycin at 37°C for 5 days.

(B) TLC assay of AFB₁ production by the WT and all mutant strains ($\Delta FKBP1$, $\Delta FKBP2$, $\Delta FKBP3$ and $\Delta FKBP4$) cultured in YES liquid medium at 29°C for 6 days.

(C) The growth inhibition rate of the WT and all mutant strains ($\Delta FKBP1$, $\Delta FKBP2$, $\Delta FKBP3$ and $\Delta FKBP4$) under rapamycin stress.

(D) AFB₁ quantitative analysis of the WT and all mutant strains ($\Delta FKBP1$, $\Delta FKBP2$, $\Delta FKBP3$ and $\Delta FKBP4$) described in (B).

(E) Phenotype of WT and all mutant strains ($\Delta FKBP3$, K5A, K19A, K40A, K42A, K55A and K65A) grown on PDA amended with 100 ng/mL rapamycin at 37°C for 5 days.

(F) The growth inhibition rate of WT and all mutant strains ($\Delta FKBP3$, K5A, K19A, K40A, K42A, K55A and K65A) under rapamycin stress.

(G) TLC assay of AFB₁ production by the WT and all mutant strains ($\Delta FKBP3$, K5A, K19A, K40A, K42A, K55A and K65A) cultured in YES liquid medium at 29°C for 6 days.

(H) Relative quantification of AFB₁ production in WT and all mutant strains ($\Delta FKBP3$, K5A, K19A, K40A, K42A, K55A and K65A) as mentioned in (G).

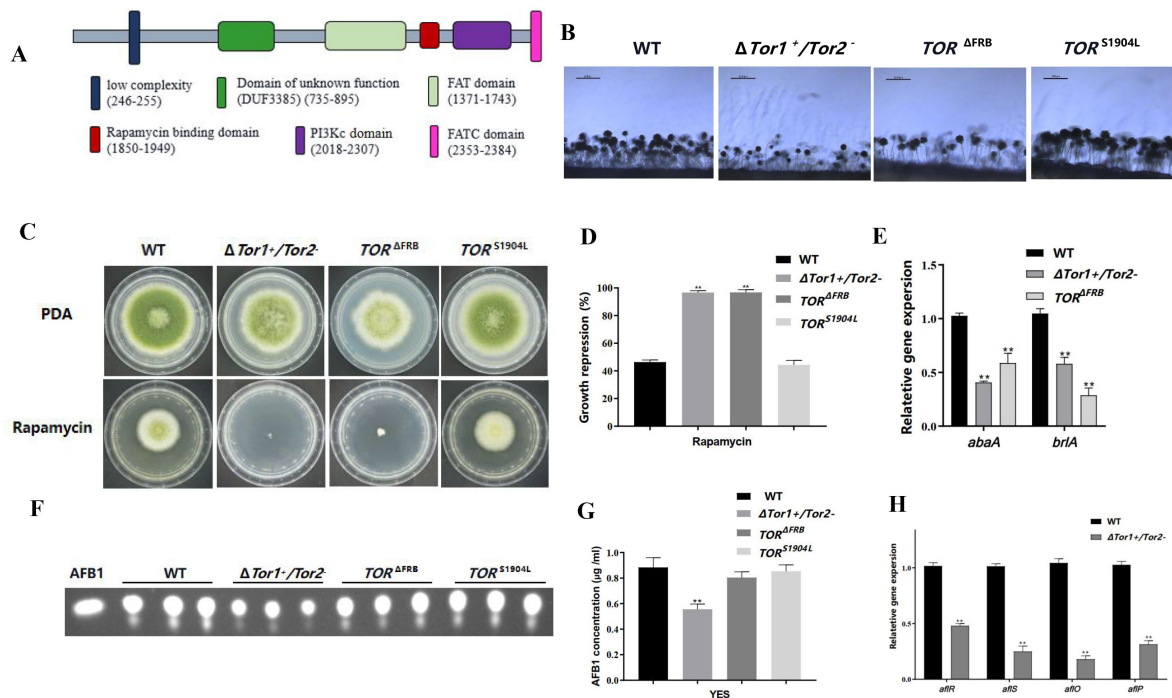


Fig. 3

Tor kinase plays important roles in *A. flavus*.

- (A) The structure diagram of Tor kinase.
- (B) Morphology of conidiophores of *A. flavus* WT, $\Delta Tor1^+/Tor2^-$, $TOR^{\Delta FRB}$ and TOR^{S1904L} strains grown on PDA medium at 37°C for 5 days as observed by microscope.
- (C) Phenotype of WT, $\Delta Tor1^+/Tor2^-$, $TOR^{\Delta FRB}$ and TOR^{S1904L} strains grown on PDA amended with 100 ng/mL rapamycin at 37°C for 5 days.
- (D) The growth inhibition rate of WT, $\Delta Tor1^+/Tor2^-$, $TOR^{\Delta FRB}$ and TOR^{S1904L} strains under rapamycin stress.
- (E) Relative transcript levels of conidia-related genes in WT, $\Delta Tor1^+/Tor2^-$, $TOR^{\Delta FRB}$ strains.
- (F) TLC assay of AFB₁ production by the WT, $\Delta Tor1^+/Tor2^-$, $TOR^{\Delta FRB}$ and TOR^{S1904L} strains cultured in YES liquid medium at 29°C for 6 days.
- (G) Quantification analysis of AFB₁ as in (F).
- (H) Relative transcript levels of aflatoxin biosynthesis regulatory and structural genes in WT and $\Delta Tor1^+/Tor2^-$ strains.

from TLC showed that AFB₁ production of the $\Delta Tor1^+/Tor2^-$ strain was dramatically decreased compared to that of WT strain (Fig. 3F). Furthermore, The qRT-PCR results showed that the expression levels of the sexual sporulation and aflatoxin synthesis genes were both downregulated in the $\Delta Tor1^+/Tor2^-$ mutant compared to WT (Fig. 3E, 3H). Interestingly, the $\Delta Tor1^+/Tor2^-$ strain cannot survive on PDA amended with 100 ng/mL rapamycin (Fig. 3C). In addition, the $\Delta Tor1^+/Tor2^-$ strain was more sensitive to osmotic stress and cell wall inhibitor, including NaCl, KCl, CFW and CR (Fig. S5). These findings suggested that TOR is involved in vegetative growth, asexual development, sclerotia formation, environmental stress and AF biosynthesis in *A. flavus*.

TOR is a conserved serine/threonine kinase, which possesses HEAT repeats at the N-terminal region, and the DUF3385, the FAT, Fkbp12-rapamycin binding (FRB), kinase catalytic and FATC domains at the C-terminal region (Fig. 3A). To assess the role of the FRB domain inside Tor kinase, the FRB domain was deleted with the method of homologous recombination. Compared with the $\Delta Tor1^+/Tor2^-$ strain, the $TOR^{\Delta FRB}$ single-copy deletion mutants exhibited similar defects in vegetative growth, conidiation, sclerotia formation, cell wall stress and rapamycin sensitivity (Fig. 3B, 3C, S4, S5). In *F. graminearum*, the S1866L mutation in the FgTor kinase confers rapamycin resistance [21]. To investigate the function of this conserved site, the TOR^{S1904L} mutant was constructed by homologous recombination, but site-specific mutation did not impair rapamycin resistance (Fig. 3C). Based on these observations, we speculated that the FRB domain plays critical roles in the regulation of Tor kinase on fungal development and rapamycin resistance in *A. flavus*.

Sch9 is involved in regulating sclerotia formation and aflatoxin biosynthesis

Sch9, a Ser/Thr kinase of the AGC family, is required for TORC1 to properly regulate ribosome biogenesis, translation initiation, and entry into the G₀ phase in *S. cerevisiae*, and TORC1-dependent phosphorylation is required for Sch9 activity [49]. The *A. flavus* putative Sch9 (AFLA_044350) gene encode a 706 amino acid protein with 80.31% identity to the *A. nidulans* Sch9. To investigate the function of Sch9 in *A. flavus*, we attempted to generate Sch9 deletion mutant using a homologous recombination strategy. The $\Delta Sch9$ strain showed normal vegetative growth and conidiation similar to the WT strain (Fig. 4B). TLC assay and quantitative analysis showed a significantly decreased aflatoxin production in $\Delta Sch9$ strain compared to WT strain (Fig. 4C). These results showed that Sch9 regulates aflatoxin biosynthesis in *A. flavus*.

Sch9 is correlate with the HOG and TOR pathways

In yeast, Sch9 is required for gene expression regulated by Sko1, which is directly targeted by the Hog1 MAP kinase, and Sch9 interacts *in vitro* with both Sko1 and Hog1 [53]. To characterize the roles of Sch9 in stress response, we examined the sensitivity of $\Delta Sch9$ to various stresses including osmotic stress and calcium stress. Compared with the WT strain, $\Delta Sch9$ significantly increased sensitivity to osmotic stress generated by NaCl and KCl (Fig. 4C). In addition, the $\Delta Sch9$ strain was more sensitive to the high concentrations of calcium stress (Fig. S5), these results indicated that the Sch9 as a element for the calcineurin-CrzA pathway in response to calcium stress. In *A. flavus*, the Sch9 kinase contain protein kinase C conserved region 2 (C2), protein kinase (S_TKc), AGC-kinase C-terminal (S_TK_X) domain and a ATP binding site (Fig. 4A). To explore the role of these domains and ATP binding site in Sch9, both domains deletion strains and $Sch9^{K340A}$ mutant were constructed. Compared with the $\Delta Sch9$ mutant, the S_TKc deletion strain ($Sch9^{\Delta S_TKc}$), S_TK_X deletion strain ($Sch9^{\Delta S_TK_X}$) and $Sch9^{K340A}$ mutant exhibited similar defects in osmotic stress and AFB₁ production (Fig. 4B-4D). Therefore, we speculate that protein kinase, kinase C-terminal domain and the ATP binding site at K340 play important roles in the regulation of Sch9 on osmotic and rapamycin resistance in *A. flavus*.

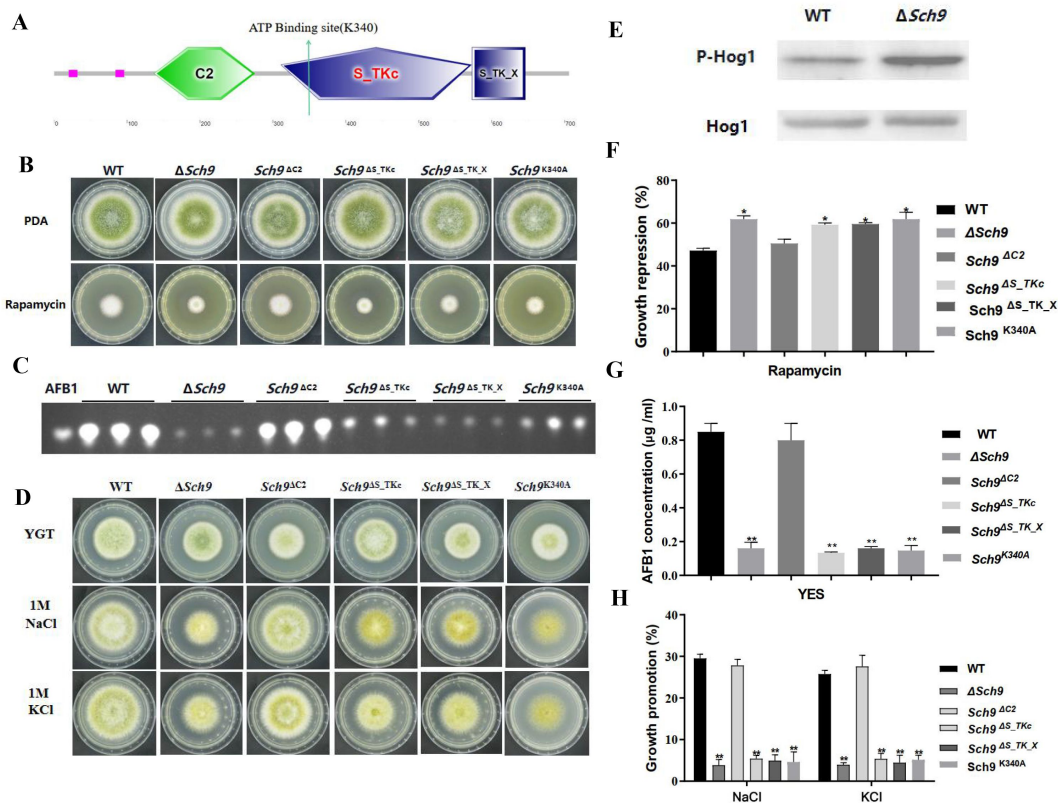


Fig. 4

Sch9 is correlated with the HOG and TOR pathways.

- (A) The structure diagram of Sch9 kinase.
- (B) Phenotype of *A. flavus* WT, $\Delta Sch9$, $Sch9^{\Delta C2}$, $Sch9^{\Delta S_TKc}$, $Sch9^{\Delta S_TK_X}$ and $Sch9^{K340A}$ strains grown on PDA amended with 100 ng/mL rapamycin at 37°C for 5 days.
- (C) TLC assay of AFB₁ production by the WT, $\Delta Sch9$, $Sch9^{\Delta C2}$, $Sch9^{\Delta S_TKc}$, $Sch9^{\Delta S_TK_X}$ and $Sch9^{K340A}$ strains cultured in YES liquid medium at 29°C for 6 days.
- (D) Colonies of WT, $\Delta Sch9$, $Sch9^{\Delta C2}$, $Sch9^{\Delta S_TKc}$, $Sch9^{\Delta S_TK_X}$ and $Sch9^{K340A}$ strains on PDA media amended with 1 M NaCl and 1 M KCl for 3 days.
- (E) The expressed levels of phospho-Hog1 were detected in WT and $\Delta Sch9$ strains.
- (F) The growth inhibition rate of WT, $\Delta Sch9$, $Sch9^{\Delta C2}$, $Sch9^{\Delta S_TKc}$, $Sch9^{\Delta S_TK_X}$ and $Sch9^{K340A}$ strains under rapamycin stress.
- (G) Quantitative analysis of AFB₁, as shown in (C).
- (H) The growth promotion rate of WT, $\Delta Sch9$, $Sch9^{\Delta C2}$, $Sch9^{\Delta S_TKc}$, $Sch9^{\Delta S_TK_X}$ and $Sch9^{K340A}$ strains under osmotic stress.

To determine if Sch9 was involved in the HOG and TOR pathways in *A. flavus*, we examined the sensitivity of $\Delta Sch9$ to rapamycin. Compared with the WT strain, the $\Delta Sch9$ strain was slightly more sensitive to rapamycin (Fig. 4B). The MAKP Hog1 is a element of the high osmolarity glycerol (HOG) pathway, so we further examined the phosphorylation level of Hog1. Western blot analysis showed that the phosphorylation level of Hog1 increased significantly in the $\Delta Sch9$ strain (Fig. 4E), indicating that Sch9 influences the Hog1 in response to osmotic stress. Taken together, these results reflected that Sch9 serves as a mediator of the TOR and HOG pathways and regulates aflatoxin biosynthesis and multiple stress responses in *A. flavus*.

TapA plays a major role in the growth, sporulation, sclerotia and aflatoxin production

Tap42, a phosphatase 2A-associating protein, is the direct target of the Tor kinase in yeast. Recent results suggested that Tap42 mediates many of the Tor functions in yeast, especially those involved in transcriptional modulation [55]. The *A. flavus* genome has a Tap42 homolog named TapA (AFLA_092770), which predicted to encode an 355 amino-acid protein shares 31.9% identity with *S. cerevisiae* Tap42. To further determine the function of the TOR signaling pathway regulator TapA in *A. flavus*, we targeted *TapA* for gene deletion. We failed to obtain a null mutant, so we generated the $\Delta TapA 1^+/TapA 2^-$ strain (containing one copy of the *TapA* gene). Compared with the WT strain, the $\Delta TapA 1^+/TapA 2^-$ strain led to reduced hyphal growth and conidiation (Fig. 5A, 5D), indicating that *TapA* is important for vegetative growth and conidiation. In addition, the $\Delta TapA 1^+/TapA 2^-$ strain was unable to produce any sclerotia (Fig. 5B), and almost lost its capability in AFB₁ production compared to the WT strain, suggesting that TapA is essential for sclerotia formation and biosynthesis of AFB₁ (Fig. 5C). The rapamycin sensitivity test showed that the $\Delta TapA 1^+/TapA 2^-$ strain exhibited decreased sensitivity to rapamycin (Fig. 5A). Furthermore, the expression of *nsdC*, *nsdD* and *sclR* were significantly decreased in $\Delta TapA 1^+/TapA 2^-$ strain compared to those in the WT (Fig. 5F). These findings revealed that the TapA plays an important positive role in growth, sporulation, sclerotia and AF biosynthesis in *A. flavus*.

TipA regulates growth, conidia, sclerotia and aflatoxin biosynthesis

In *S. cerevisiae*, Tip41 was identified as a Tap42-interacting protein and has been measured to function as a negative regulator of Tap42, which downregulates TORC1 signaling via activation of PP2A phosphatase [35]. The putative Tip41 orthologue in *A. flavus* named TipA (AFLA_047310). The amino acid sequence of TipA shares 40.49% similarity to *S. cerevisiae* Tap41. To elucidate the function of TOR signaling pathway protein TipA in *A. flavus*, we generated the TipA deletion mutant. The $\Delta TipA$ strain exhibited smaller diameter colonies on PDA compared to the WT strain, and the number of conidia produced by $\Delta TipA$ strain was significantly reduced than that by the WT strain on PDA (Fig. 5A). In addition, the $\Delta TipA$ strain was unable to produce any sclerotia (Fig. 5B), and TLC analyses indicated that the deletion of TipA led to a considerable decrease in AFB₁ production on YES (Fig. 5C). Meanwhile, the $\Delta TipA$ strain exhibited decreased sensitivity to rapamycin (Fig. 5A). Moreover, the expression levels of genes *nsdC* and *nsdD* showed a decrease in $\Delta TipA$ strain compared to WT (Fig. 5F). These results indicated that TipA is important for growth, conidia, sclerotia and aflatoxin biosynthesis in *A. flavus*.

Phosphatase SitA and Ppg1 are involved in growth, conidiation and sclerotia formation

In *S. cerevisiae*, Tap42 also interacts with the catalytic subunits of Type 2A phosphatases, such as Pph21, Pph22, Pph3, Sit4 and Ppg1, which is controlled by the Tor signaling pathway [36]. Phosphatase SitA (AFLA_108450) and Ppg1 (AFLA_132610) were identified in *A. flavus*, which encode putative orthologs of Sit4 and Ppg1 in *S. cerevisiae*. To determine the biological functions of

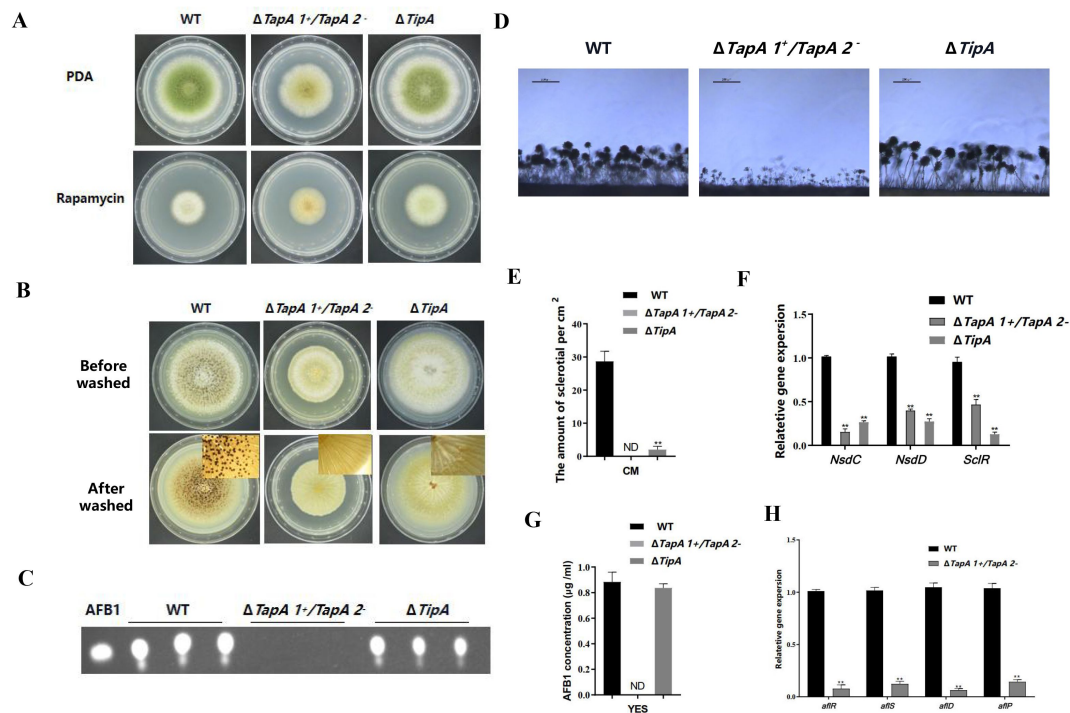


Fig. 5

TapA and TipA regulate growth, sclerotia and aflatoxin biosynthesis in *A. flavus*.

(A) Phenotype of the *A. flavus* WT, $\Delta TapA 1^+/TapA 2^-$ and $\Delta TipA$ strains grown on PDA amended with 100 ng/mL rapamycin at 37°C for 5 days.

(B) Colony phenotype of the WT, $\Delta TapA 1^+/TapA 2^-$ and $\Delta TipA$ strains grown on CM medium at 37°C for 7 days.

(C) TLC assay of AFB₁ production by the WT, $\Delta TapA 1^+/TapA 2^-$ and $\Delta TipA$ strains cultured in YES liquid medium at 29°C for 6 days.

(D) Fluorescent images of hyphal cells for the WT, $\Delta TapA 1^+/TapA 2^-$ and $\Delta TipA$ strains stained with calcofluor white (CFW).

(E) Amounts of sclerotia produced by the WT, $\Delta TapA 1^+/TapA 2^-$ and $\Delta TipA$ strains.

(F) Relative transcript levels of sclerotia-related genes in WT and $\Delta TapA 1^+/TapA 2^-$ strains.

(G) Relative quantification of AFB₁ production as in (C).

(H) Relative transcript levels of aflatoxin biosynthesis regulatory and structural genes in WT and $\Delta TapA 1^+/TapA 2^-$ strains.

phosphatase SitA and Ppg1 in *A. flavus*, we disrupted the *SitA* and *Ppg1* by homologous recombination strategy. To further study the relationship between phosphatase SitA and Ppg1, we also constructed complement and double deletion mutants. The vegetative development phenotype analysis showed that the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* strains grew significantly more slowly than the WT strain on PDA (Fig. 6A). In addition, the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* strains formed lower number of conidiophores (Fig. 6A). To gain further insight into the role of SitA and Ppg1 in conidiation, qRT-PCR was used to determine the transcript levels of conidial synthesis-related genes (*abaA* and *brlA*). The expression levels of the *abaA* and *brlA* genes were significantly downregulated in the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* mutants compared to WT and complemented strains (Fig. 6G). The sclerotial formation analysis results showed that the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* mutants were unable to produce any sclerotia (Fig. 6B). To further prove these findings, we conducted qRT-PCR to determine the expression levels of three regulators (*NsdC*, *NsdD* and *SclR*) of sclerotia development. Our results showed that the expression levels of *NsdC*, *NsdD* and *SclR* were all downregulated in the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* mutants (Fig. 6H), which is consistent with the sclerotia production observed in the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1*. These results showed that SitA and Ppg1 are crucial for hyphal development, conidiation and sclerotia formation in *A. flavus*.

SitA and Ppg1 play different roles in regulating aflatoxin biosynthesis

Previous studies have demonstrated that protein phosphorylation is critical for the regulation of AFs biosynthesis, which are the toxic and carcinogenic secondary metabolites of *A. flavus* [37]. We assayed AFB₁ biosynthesis in the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* mutants, and TLC analyses indicated that Δ *SitA* and Δ *SitA*/ Δ *Ppg1* strains produced less AFB₁, while the level of AFB₁ produced by Δ *Ppg1* was markedly increased compared to that of the WT and complemented strains (Fig. 6C). The expression levels of aflatoxin synthesis-related genes (structural genes: *aflO*, *aflP*, *aflQ* and *aflJ*, and regulatory genes: *aflR* and *aflS*) were determined by qRT-PCR in the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* mutants, and the result showed that the expression levels of those genes were drastically downregulated in the Δ *SitA* and Δ *SitA*/ Δ *Ppg1* mutants, while upregulated in Δ *Ppg1* mutants when compared to those of the WT and complemented strains (Fig. 6I). These results indicated that SitA and Ppg1 play different roles in regulating aflatoxin biosynthesis in *A. flavus*.

SitA and Ppg1 are important for pathogenicity

To investigate the pathogenicity of phosphatases SitA and Ppg1 on crops, we conducted pathogenicity tests on peanut and maize. The results showed that WT and complement strains resulted in full virulence on all peanut and maize seeds, while the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* mutants exhibited more flawed colonization on peanut and maize seeds (Fig. 7A). In addition, the number of conidial productions of these mutants on the infected peanut seeds also dramatically decreased compared to those of WT and complement strains (Fig. 7C). TLC analyses indicated that the Δ *Ppg1* mutant produced more AFB₁ on peanut and maize, while the level of aflatoxin produced by Δ *SitA* and Δ *SitA*/ Δ *Ppg1* mutants on peanut and maize were significantly reduced compared to WT and complement strains (Fig. 7B and 7D). This is consistent with the above results of aflatoxin biosynthesis of all mutants in the medium. All these data showed that SitA and Ppg1 are important for crop seeds pathogenicity.

SitA and Ppg1 are involved in regulating cell wall integrity

To explore the role of phosphatases SitA and Ppg1 in response to cell wall stress, all strains were cultured on PDA amended with CFW and CR. We found that the relative growth inhibition of Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* mutants induced by cell wall stress were significantly increased compared to WT and complement strains (Fig. 8A and 8B). Further studies by qRT-PCR showed that expression levels of genes associated with synthetic cell walls, such as chitin synthase genes *chsA*,

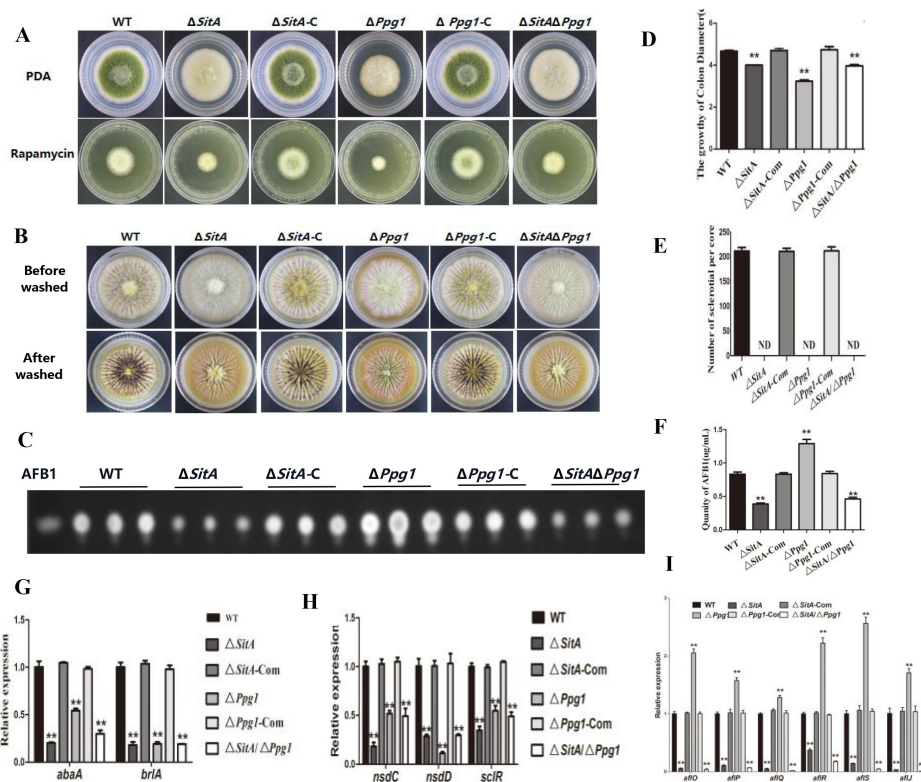


Fig. 6

Phosphatases SitA and Ppg1 are involved in growth, conidiation and aflatoxin biosynthesis in *A. flavus*.

- (A) The colony morphology of *A. flavus* WT, single knockout and double knockout strains of *SitA* and *Ppg1* cultured on PDA medium amended with 100 ng/mL rapamycin at 37°C for 5 days.
- (B) colony morphology of *A. flavus* WT, single knockout strain and double knockout strain of *SitA* and *Ppg1* cultured on WKM medium at 37°C for 7 days.
- (C) TLC detection of WT, single knockout and double knockout strains of *SitA* and *Ppg1* cultured in YES liquid medium at 29°C for 6 days.
- (D) Comparison of conidial production among WT, single knockout and double knockout strains on PDA media.
- (E) Sclerotia statistics of WT, single knockout and double knockout strains.
- (F) Quantitative analysis of AFB₁ of WT, single knockout and double knockout strains.
- (G) Relative transcript levels of conidia synthesis genes in WT, single knockout and double knockout strains.
- (H) Relative transcript levels of sclerotia synthesis genes in WT, single knockout and double knockout strains.
- (I) Relative transcript levels of aflatoxin biosynthesis genes in WT, single knockout and double knockout strains.

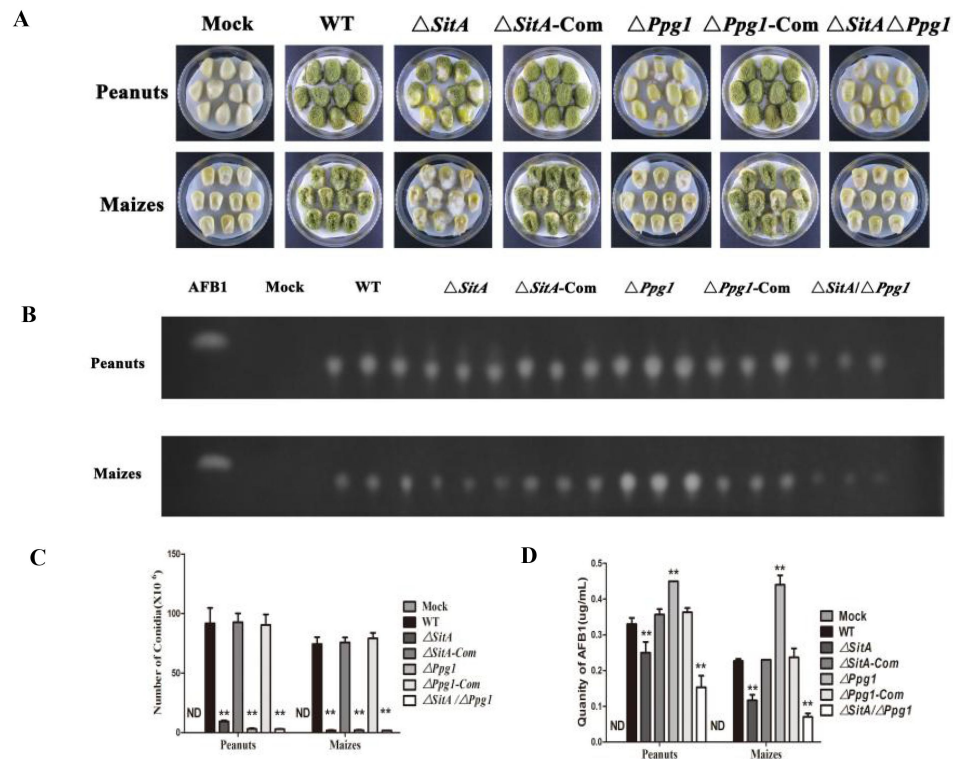


Fig. 7

Phosphatase *SitA* and *Ppg1* are important for pathogenicity in *A. flavus*.

- (A) Phenotypes of peanut and maize infected by *A. flavus* WT, single and double knockout strains of *SitA* and *Ppg1*.
 (B) TLC detection of toxin from peanut and maize infected by WT, single knockout and double knockout strains.
 (C) Spores statistics from peanut and maize infected by WT, single and double knockout strains.
 (D) Quantitative analysis of AFB₁ from peanut and maize infected by WT, single knockout and double knockout st

chsB, *chsC* and *chsD*, were significantly decreased in the Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* strains (**Fig. 8C**). Western blot analysis showed that the phosphorylation level of Slt2 was higher in Δ *SitA*, Δ *Ppg1* and Δ *SitA*/ Δ *Ppg1* than that in the WT strain (**Fig. 8D**). To sum up, the phosphatases SitA and Ppg1 maintained the integrity of the cell wall by positively regulating the expression of those genes.

The phosphatase complex Nem1/Spo7 are important for growth, conidiation, aflatoxin and LD biogenesis

Lipid droplets (LDs) are spherical organelles, which appear in the cytosol of some prokaryotic and most eukaryotic cells upon energy surplus. The FgTOR pathway regulates the phosphorylation status of FgNem1 mainly via the FgPpg1/Sit4-FgCak1/other kinase cascade in *F. graminearum* [7]. In this study, we observed that rapamycin-treated hyphae led to the accumulation of LD biogenesis (**Fig. 9A**), suggesting that the TOR pathway may regulate the LD biogenesis. However, the functions of LDs and the underlying regulatory mechanism of LD biosynthesis are still largely unknown. In order to explore the regulatory mechanism of the TOR pathway that regulates LD biogenesis in *A. flavus*, we examined LD accumulation in Δ *Sch9*, Δ *SitA* and Δ *Ppg1*. BODIPY staining assays showed that LD biogenesis was not induced in the Δ *SitA* and Δ *Ppg1* strains treated with rapamycin compared to those in the WT and Δ *Sch9* strains (**Fig. 9A-D**). These results indicated that LD biogenesis induced by rapamycin is largely dependent on the phosphatases SitA and Ppg1 in *A. flavus*. In *F. graminearum*, the phosphatase FgNem1/FgSpo7 complex was involved in induction of LD biogenesis by rapamycin. Phosphatase Nem1(AFLA_002649) and Spo7 (AFLA_028590) were identified in *A. flavus*, which encode putative orthologs of FgNem1 and FgSpo7 in *F. graminearum*. To investigate the function of Nem1 and Spo7 gene in *A. flavus*, Δ *Nem1* and Δ *Spo7* mutants was constructed with homologous recombination. The Δ *Nem1* and Δ *Spo7* mutants exhibited a significant decrease in vegetative growth and conidiation compared to WT (**Fig. 10A**). TLC assay and quantitative analysis showed a significantly decreased aflatoxin production in Δ *Nem1* and Δ *Spo7* compared to WT (**Fig. 10B**). In addition, BODIPY staining showed no visible lipid droplets in the hyphae of Δ *Nem1* and Δ *Spo7* after rapamycin treatment (**Fig. 10C**). Taken together, these results indicated that Nem1/Spo7 are involved in vegetative growth, conidiation, aflatoxin and LD biogenesis.

Discussion

Aflatoxins (AFs) are a class of toxic secondary metabolites produced by *A. flavus*, including aflatoxin B₁, B₂, G₁ and G₂, among which AFB₁ has been regarded as the most pathogenic and toxic aflatoxin [38]. Therefore, understanding the regulatory mechanism of fungal secondary metabolite biosynthesis would help in exploiting aflatoxin control in *A. flavus*. In this study, we demonstrated that the target of the rapamycin (TOR) signaling pathway plays critical roles in regulating growth, sporulation, sclerotia and aflatoxin production in *A. flavus*. Through functional studies of nine genes in the TOR signaling pathway, we explored the conservation and complexity of the TOR signaling pathway in *A. flavus*. In addition, we constructed a crosstalk network between TOR and other signaling pathways (MAPK, CWI) and analyzed the specific regulatory process. We proposed that the TOR signaling pathway interacts with other signaling pathways (MAPK, CWI, calcineurin-CrzA pathway) to regulate the responses to various environmental stresses (**Fig. 11**). Our results might provide new insights into the biological control of *A. flavus*.

We found that rapamycin not only affects the growth and sporulation of *A. flavus*, but also strongly inhibits sclerotia production and aflatoxin biosynthesis (**Fig. 1A-1C**). This inhibitory effect has been observed in a variety of fungi, including *Candida albicans* [39], *Cryptococcus neoformans* [39], *Podospira anserine* [40], *Aspergillus nidulans* [26], *Magnaporthe oryzae* [8], and *Fusarium graminearum* [21], suggesting that the function of the TOR signaling pathway may be

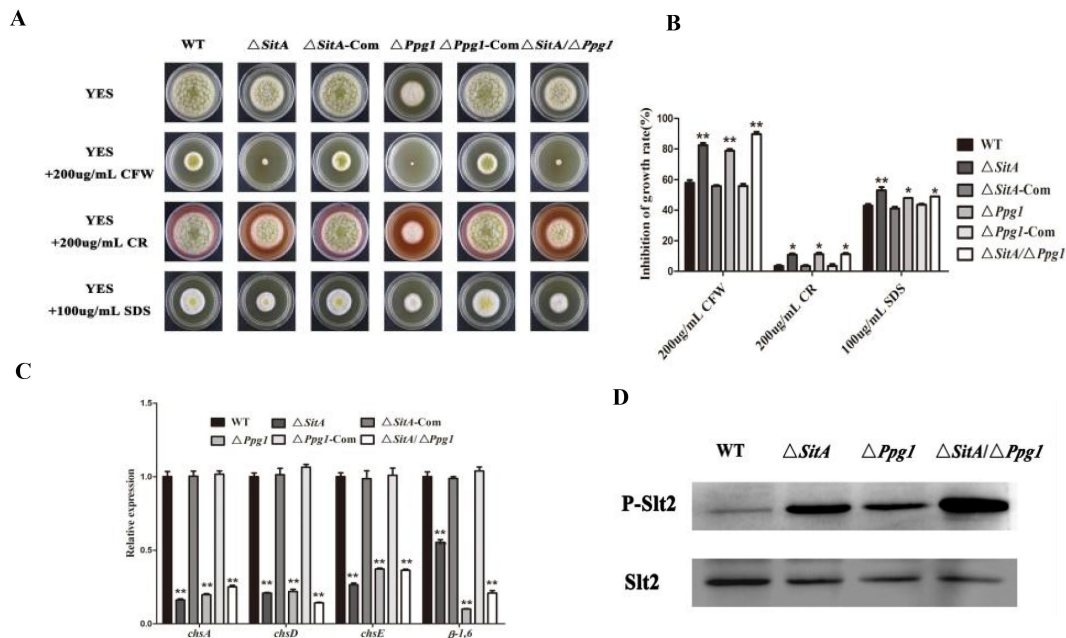


Fig. 8

Phosphatase SitA and Ppg1 are involved in regulating cell wall integrity.

- (A) Morphology of *A. flavus* WT, single knockout and double knockout strains of *SitA* and *Ppg1* grown on PDA media supplemented with 200 μ g/mL CR, 200 μ g/mL CFW, or 100 μ g/mL SDS at 37°C for 5 days.
- (B) The growth inhibition rate of the WT, single knockout and double knockout strains under cell wall and cell membrane stress.
- (C) Expression levels of cell wall synthesis genes (*chsA*, *chsD*, *chsE* and β -1,6) in WT, single knockout and double knockout strains.
- (D) Phosphorylation level of Slr2 in WT, single knockout and double knockout strains was detected by Western blotting.

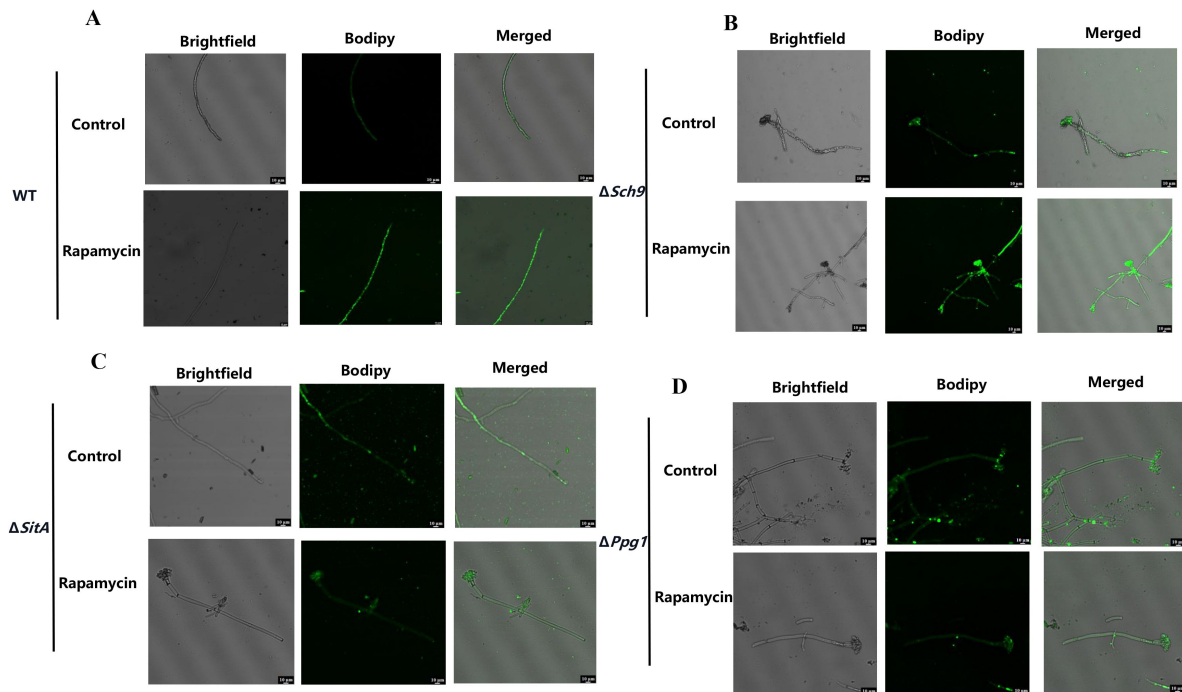


Fig. 9

Phosphatase SitA and Ppg1 are involved in regulating lipid droplet biogenesis.

(A) The intracellular lipid droplets were stained by boron dipyrromethene difluoride (BODIPY) in hyphae of the *A. flavus* WT strain and observed under a fluorescence microscopy. The Phenotype of lipid droplets accumulation in the mycelia of *A. flavus* WT strain treated with or without 100 ng/mL rapamycin cultured for 6 h. Bar, 10 μ m.

(B) The phenotype of lipid droplets accumulation in the mycelia of $\Delta Sch9$ strain treated with or without 100 ng/mL rapamycin cultured for 6 h. Bar, 10 μ m.

(C) The phenotype of lipid droplets accumulation in the mycelia of $\Delta SitA$ strain treated with or without 100 ng/mL rapamycin cultured for 6 h. Bar, 10 μ m.

(D) The phenotype of lipid droplets accumulation in the mycelia of $\Delta Ppg1$ strain treated with or without 100 ng/mL rapamycin cultured for 6 h. Bar, 10 μ m.

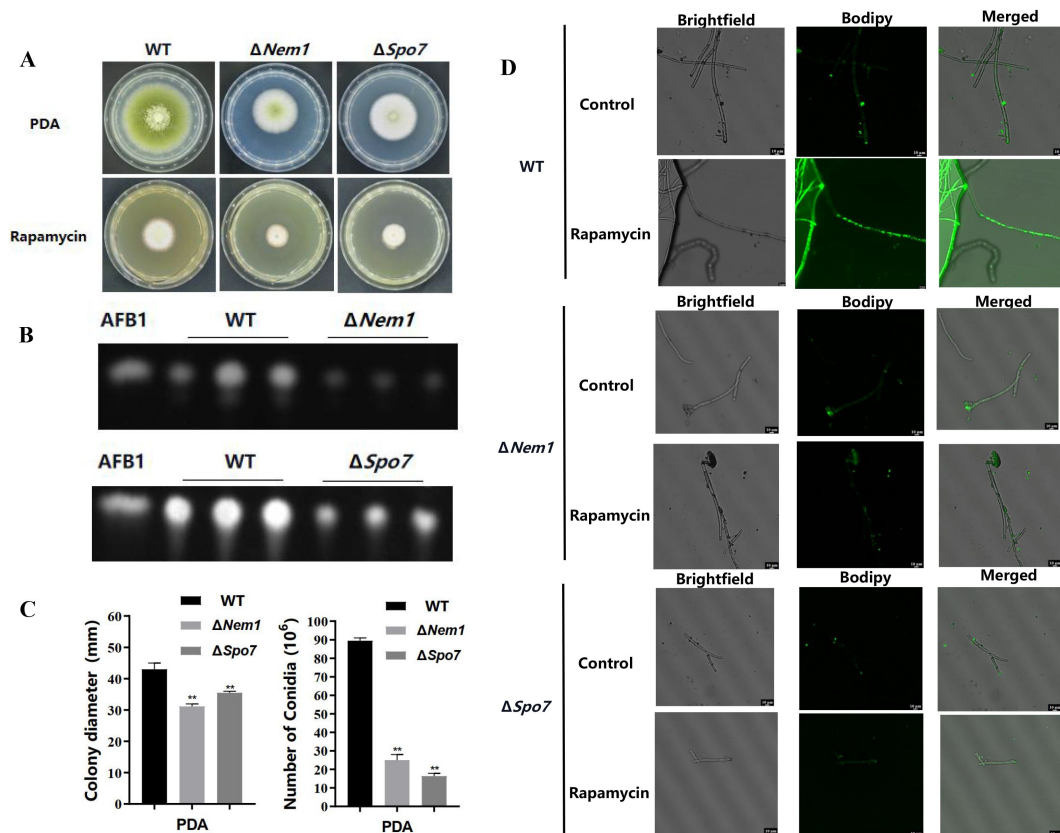


Fig. 10

Phosphatase complex Nem1/Spo7 play an important role in growth, conidiation and aflatoxin and lipid droplet biogenesis.

- (A) Phenotype of the *A. flavus* WT, $\Delta Nem1$ and $\Delta Spo7$ strains grown on PDA amended with 100 ng/mL rapamycin at 37°C for 5 days.
- (B) TLC assay of AFB₁ production by the *A. flavus* WT, $\Delta Nem1$ and $\Delta Spo7$ strains cultured in YES liquid medium at 29°C for 6 days.
- (C) Statistical analysis of the diameter and sporulation by the *A. flavus* WT, $\Delta Nem1$ and $\Delta Spo7$ strains.
- (D) The Phenotype of lipid droplets accumulation in the mycelia of the *A. flavus* WT, $\Delta Nem1$ and $\Delta Spo7$ strains treated with or without 100 ng/mL rapamycin cultured for 6 h. Bar, 10 μ m.

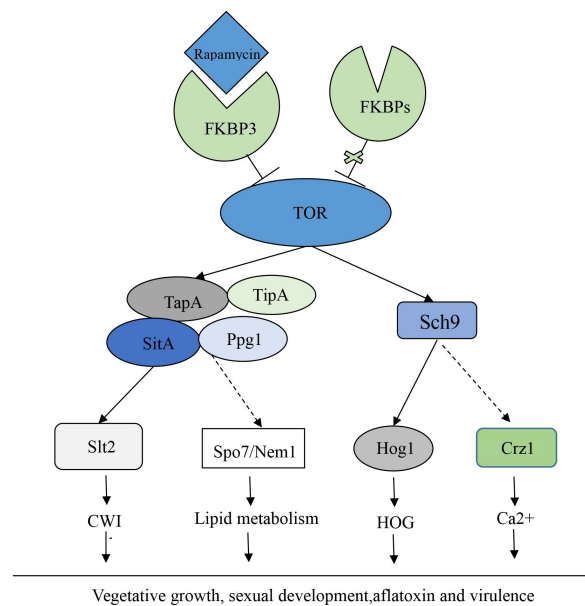


Fig. 11

The proposed model of target of the rapamycin (TOR) pathway in *A. flavus*.

Based on the above results, we hypothesized the TOR signaling pathway model in *A. flavus*. Rapamycin forms a complex with FKBP3, and this complex can bind to the Tor kinase to inhibit its normal function. Sch9 as a downstream component of the Tor kinase, regulates sclerotia formation, aflatoxin biosynthesis, the TOR and HOG signaling pathways. In addition, We speculate that the Sch9 as a element for the calcineurin-CrzA pathway in response to calcium stress. As another important target of the Tor kinase, Tap42-phosphatase complex is involved in the regulation of growth, sporulation, sclerotia, aflatoxin production, CWI signaling pathways, lipid droplet synthesis and other processes. In conclusion, the TOR signaling pathway plays an important role in the vegetative development, stress responses, aflatoxin biosynthesis and pathogenicity in *A. flavus*.

conserved in filamentous fungi. The number of FK506-binding proteins varies among species: there are four FKBP proteins in *S. cerevisiae* and three FKBP proteins in *S. pombe*^[34]. In *S. cerevisiae*, rapamycin can bind to FKBP protein, irreversibly inhibit the G₁ phase of the cell cycle and control cell growth^[46]. The entomopathogenic fungus *Beauveria bassiana* contains three putative FKBP genes, and disruption of *BbFKBP12* significantly increases the resistance to rapamycin and FK506^[41]. The plant pathogenic fungus *F. graminearum* harbors three FKBP proteins, among which deletion of *FgFKBP12* leads to resistance to rapamycin^[21]. In this study, we showed that FKBP3 is involved in sclerotia and aflatoxin biosynthesis and rapamycin resistance in *A. flavus*(**Fig. 2A-2C**), and the conserved site K19 of FKBP3 regulated the aflatoxin biosynthesis and rapamycin resistance(**Fig. 2E-2G**). Previous research has shown that FKBP12 was characterized as the receptor of FK506 and rapamycin, which has been the most extensively documented in different fungi. The deletion of FKBP12 confers the resistance to rapamycin in *C. albicans*^[39] and *C. neoformans*^[39]. Similar phenotypes are observed in other fungi, such as *F. graminearum*^[21], *B. bassiana*^[41] and *Mucor circinelloides*^[30], and the FKBP12 mutants have significantly increased resistance to rapamycin and FK506 in these filamentous fungi. In *Botrytis cinerea*^[28], the deletion of *BcFKBP12* cause a reduction of the virulence of strain T4, while knockout of the FKBP12 gene did not affect the pathogenic development of the strain B05.10. Thus, we proposed that these FKBP proteins may perform similar functions in fungi.

The Tor kinase acts as a core element of the TOR signaling pathway, which is evolutionarily conserved in fungi, plants and mammals^[45]. TOR assembles into two distinct multiprotein complexes: TORC1 and TORC2. A defining feature of TORC1 is the interaction of TOR with KOG1 (Raptor in mammals) and its sensitivity to a rapamycin-FKBP12 complex^[61]. Tor kinase is widely conserved both structurally and as the target of FKBP-rapamycin. Almost all other eukaryotes, including plants, worms, flies, and mammals, have a single Tor gene, whereas *S. cerevisiae* has two Tor genes^[47]. Tor kinase is widely conserved both structurally and as the target of FKBP-rapamycin, which have similar domains including HEAT-like, FAT, FRB, kinase and FATC domains. The FRB domain is the binding region of the FKBP-Rapamycin complex, and rapamycin resistance is caused by the disruption of the FRB domain^[47]. The homology comparison and phylogenetic analysis revealed that there is the only Tor kinase gene predicted in *A. flavus*. To determine its function, we generated the *Tor* single-copy deletion mutant by homologous recombination. We found that Tor was involved in the vegetative growth and asexual development of *A. flavus*(**Fig. 3C**). In addition, Tor responds to various stresses including osmotic and oxidative stresses, cell wall-damaging agents and rapamycin(**Fig. 3F**, **S6**). Our results revealed that Tor kinase and its FRB domain play very important roles in the morphogenesis, mycotoxin biosynthesis, and multiple stress responses of *A. flavus*. Previous findings demonstrated that Tor kinase could function as a molecular switch connecting an iron cue to defend against pathogen infection in *C. elegans*^[57]. In *A. fumigatus*, Tor kinase represents a central regulatory node controlling genes and proteins involved in nutrient sensing, stress response, cell cycle progression, protein biosynthesis and degradation, as well as adaptation to low iron conditions^[58]. In *F. graminearum*, the Tor point mutation (S1866L) prevents Fkbp12-rapamycin complex from binding to the FgFRB domain of FgTor^[21]. We also constructed a point mutant strain *TOR*^{S1904L}, but the sensitivity to rapamycin is consistent with the WT strain, which may be related to the functional variances of the Tor site (S1866L) in different species.

In *S. cerevisiae*, the protein kinase Sch9 is a major downstream effector of TORC1, which plays important roles in stress resistance, longevity and nutrient sensing^[59]. In addition, the TORC1-Sch9 pathway acts as a crucial mediator of chronological lifespan^[50]. In filamentous fungi, there is a complex network of interactions between TOR and HOG pathways. In *F. graminearum*^[7], FgSch9 and FgHog1 mutants exhibited increased sensitivity to osmotic and oxidative stresses, and this defect was more severe in the FgSch9/FgHog1 double mutant. The phosphorylation level of FgHog1 increased significantly in Δ FgSch9, and the affinity capture-MS assay showed that TOR was one of the proteins co-purifying with FgSch9. These results suggest that FgSch9 serves as a mediator for the TOR and HOG pathways^[7]. In addition, FgSch9 regulates hyphal differentiation,

asexual development, virulence and DON biosynthesis, which is also involved in the regulation of sensitivity to rapamycin, cell wall-damaging agents, fungicide, osmotic and oxidative stresses^[7]. In *A. fumigatus*, the *SchA* mutant was sensitive to rapamycin, high concentrations of calcium and hyperosmotic stress, and *SchA* was involved in iron metabolism. The *SchA* mutant showed increased phosphorylation of SakA upon osmotic stress^[51]. In this study, we discovered that *Sch9* is involved in regulating aflatoxin biosynthesis (**Fig. 4C**), and responses to various stresses including osmotic stress, cell wall-damaging agents and rapamycin (**Fig. 4B**). The S_TKc domain deletion strain (*Sch9*^{ΔS_TKc}), S_TK_X domain deletion strain (*Sch9*^{ΔS_TK_X}) and *Sch9*^{K340A} mutant exhibited similar phenotype with *ΔSch9* strain (**Fig. 4C-4D**). These findings revealed that *Sch9* plays an important role in the biological synthesis of aflatoxin and multiple stresses mainly via its kinase domain, kinase-C domain and ATP binding site k340. We also found that the phosphorylation level of Hog1 increased significantly in the *ΔSch9* strain (**Fig. 4E**). Based on the above results, we speculated that *Sch9* serves as a mediator for the complex network of interactions amongst the TOR and HOG signaling pathways.

In *S. pombe*, Tip41 (a Tap42-interacting protein) plays an important role in the cellular responses to nitrogen sources by regulating type 2A phosphatases^[42]. In *S. cerevisiae*, Tap42-phosphatase complexes associate with TORC1, and Tip41 downregulates TORC1 signaling via activation of PP2A phosphatase^[34]. Unlike yeast Tip41, TIPRL, a mammalian Tip41-like protein, has a positive effect on mTORC1 signaling through the interaction with PP2Ac^[43]. While the interaction between Tip41 and Tap42 was not observed in the filamentous fungi. The Y2H assay showed that the FgTip41 interacts with phosphatase FgPpg1 rather than FgTap42, while the FgTap42 bind with phosphatases FgPp2A, implying that FgTap42, FgPpg1 and FgTip41 form a heterotrimer in *F. graminearum*^[21]. FgTip41 is associated with mycotoxin production and pathogenicity in *F. graminearum*, but *ΔFgTip41* had no detectable changes in sensitivity to rapamycin^[21]. In the rice blast fungus *M. oryzae*, MoTip41 does not interact with the MoTap42, but MoTip41 can bind with phosphatase MoPpe1 to mediate crosstalk between TOR and cell wall integrity signaling^[9]. In this study, we found that the TapA and TipA were essential for growth, conidia, sclerotia and aflatoxin biosynthesis in *A. flavus* (**Fig. 5A-5C**). The *TapA* single-copy deletion mutant exhibited decreased sensitivity to rapamycin. However, the *TipA* deletion mutant had no detectable changes in sensitivity to rapamycin. These results indicated that the functions of TapA and TipA in *A. flavus* were similar to its ortholog in the filamentous fungi.

In *S. cerevisiae*, Tap42 can bind and regulate all PP2A phosphatases, including Pph3, Pph21, Pph22, Sit4 and Ppg1, participating in the TOR signaling pathway in response to rapamycin treatment or nutrient deprivation^[35]. Sit4 plays an important role in cell growth and environmental stimuli responses, such as high glucose environment, high temperature and hygromycin^[44]. FgSit4 and FgPpg1 are associated with various cellular processes, including mycelial growth, conidiation and DON biosynthesis virulence in *F. graminearum*^[21]. In addition, FgSit4 and FgPpg1 are involved in the regulation of various signaling pathways, such as the TOR signaling pathway, CWI pathway and lipid droplets biogenesis regulation pathway^[22]. In this study, we found that the phosphatase SitA and Ppg1 play critical role in vegetative growth, conidiation, pathogenicity, LD biogenesis, sclerotia formation and aflatoxin biosynthesis in *A. flavus* (**Fig. 6A-6C**). Additionally, SitA and Ppg1 also affect cell wall integrity stresses response and rapamycin sensitivity. After the deletion of the *SitA* and *Ppg1* genes, the expression level of genes related to cell wall synthesis has decreased significantly, resulting in damage to the cell wall and showing a higher sensitivity to cell wall integrity stress. Western blotting results showed that the phosphorylation level of MAPK kinase Slt2 has increased significantly in the *ΔSitA* and *ΔPpg1* strains (**Fig. 8D**), indicating that SitA and Ppg1 have a negative regulation on the CWI pathway. We concluded that phosphatases SitA and Ppg1 serve as a mediator for the TOR and CWI pathways in *A. flavus*. In *M. oryzae*, MoPpe1 is the homologue of *S. cerevisiae* serine/threonine protein phosphatase Sit4/Ppe1, which is required for vegetative growth, conidiation and full virulence. MoPpe1 and MoSap1 function as an adaptor complex linking CWI and TOR signaling and the activation of the TOR pathway leads to suppression of CWI signaling^[8]. In addition, the phosphatase MoPpe1 interacts with

phosphatase-associated protein MoTip41 to mediate crosstalk between TOR and CWI signaling^[8]. These results suggest that phosphatase ppe1 interacts with different proteins to coordinate the TOR and CWI signaling pathways in regulating the growth and pathogenicity of rice blast fungus *M. oryzae*. Taken together, we speculated that phosphatase SitA and Ppg1 may regulate CWI and TOR pathways differently between species to perform their function.

In *F. graminearum*, the phosphatase complex FgNem1/Spo7-FgPah1 cascade plays important roles in fungal development, DON production and LD biogenesis^[22]. In this study, our results indicated that Nem1 and Spo7 are important for asexual development and secondary metabolism of *A. flavus*. We also found that LD biogenesis mediated by the Nem1/Spo7 and SitA/Ppg1 in *A. flavus*. These observations imply the PP2A phosphatases are core phosphatase in the LD biogenesis signaling network, there should be other kinases involved in regulating the LD biogenesis. It seems that phosphatases may be related to cross-talking among the TOR and LD biogenesis signaling network. The exact molecular mechanism is required to investigate the relationship between phosphatase and other kinase and to explore regulatory mechanism and biological functions of the LDs.

In conclusion, we identified the complex regulatory network that elucidates how crosstalk between TOR, HOG and CWI signaling modulates the development, aflatoxin biosynthesis and pathogenicity of *A. flavus*. By resolving the functions of genes related to the TOR signaling pathway, our findings can lay a solid foundation for an in-depth understanding of the molecular mechanism of *A. flavus* pathogenicity, which would provide novel insight for controlling AF contamination and reducing *A. flavus* infection.

Materials and methods

Strains and cultural conditions

The *A. flavus* strains in this study were shown in S1 Table. All strains were cultured on potato dextrose agar (PDA), yeast extract-sucrose agar (YES) and yeast extract-glucose agar (YGT) at 37°C for growth and conidiation assays respectively, and YES liquid medium and Potato dextrose broth (PDB) were used to detect aflatoxin production at 29°C.

Constructions of gene deletion and complementation mutants

The gene deletion and complementation strains were constructed using the SOE-PCR strategy according to previously described methods^[32]. The primers used to amplify the flanking sequences of each gene were listed in S2 Table. The putative gene deletion mutants were verified by PCR and RT-PCR assays with the relevant primers (Fig S1) and further confirmed by sequencing analysis.

Growth, conidiation and sclerotia analysis

To assess mycelial growth and conidia formation, 1 µL of 10⁶ spores/mL conidia of all strains were point inoculated onto the center of YES, PDA and YGT agar medium respectively, and then cultured at 37°C for 5 days in the dark. The colony diameters were measured every day, and the conidia were washed by 0.07% Tween-20 and counted using a hemocytometer and microscope^[37]. For sclerotia formation analysis, each strain was inoculated and grown on CM agar medium at 37°C in the dark for 7 days, and then 75% ethanol was used to wash away mycelia and conidia on the surface of the medium^[60]. The assays were repeated at least three times.

Extraction and quantitative analysis of aflatoxin

To extract AF and quantify the AF production, 1 μL of 10^6 spores/mL conidia of all strains were point inoculated onto liquid YES or PDB medium and cultures were incubated at 29°C in the dark for 6 days. AF were extracted from the media and detected by thin layer chromatography (TLC) [60]. In addition, the standard AFB₁ was used for visual comparison, and AF was quantitative analyzed by the Image J tool. The assays were repeated at least three times.

Stress assay

To determine the roles of the gene in *A. flavus* responding to various environmental stresses, 1 μL of 10^6 spores/mL conidia of all strains were point inoculated onto these PDA solid plates supplemented with different agents, including 1M NaCl, 1M KCl and 1M CaCl₂ treatment for osmotic stress, 100 $\mu\text{g/mL}$ SDS, 200 $\mu\text{g/mL}$ Calcofluor white (CFW) and 200 $\mu\text{g/mL}$ Congo red (CR) treatment for cell wall stress. After 5 days of incubation at 37°C in the dark, the relative inhibition rates were calculated [37]. The assays were repeated at least three times.

Seeds infection assays

To measure the pathogenicity of *A. flavus*, all peanut and maize seeds were washed in 0.05% sodium hypochlorite, 75% ethanol and sterile water respectively, and then all peanut and maize seeds were soaked in sterile water supplemented with 10^6 spores/mL of all strains at 29°C in the dark for 60 min. After the conidia adhesion, the seeds were cultured at 29°C for 6 days in the dark [60].

Quantitative RT-PCR

For real-time quantitative PCR (qPCR) assays, the *A. flavus* strains were cultured in PDA medium at 29°C for 3 days in the dark. Total RNA was extracted from the mycelia of all strains, and cDNA was synthesized with First Strand cDNA Synthesis Kit, and then the cDNA was used as the template for qRT-PCR analysis with SYBR Green qPCR mix. The relative transcript level of each gene was calculated by the $2^{-\Delta\Delta\text{CT}}$ method [62].

Western blot analysis

To extract the total protein, the mycelia of each strain were cultured in YES liquid medium for 48 h, then the sample was grinded under liquid nitrogen to a fine powder and transferred to a 1.5 mL microcentrifuge tube added with RIPA lysis buffer and 1 mM PMSF protease inhibitor. Proteins were separated by 12% SDS-PAGE and transferred onto a polyvinylidene fluoride (PVDF) membrane. Phosphorylation levels of Hog1 and Slt2 were detected by Phospho-p38 MAPK antibody (Cell Signaling Technology, Boston, MA, USA) and Phospho-p44/42 MAPK (Erk 1/2) antibody (Cell signaling Technology, MA, USA), respectively [60]. The experiment was conducted three times independently.

Statistical analysis

All data were measured with means $\pm$ standard deviation (SD) of three biological replicates. Graph Pad Prism 7 software was used for statistical and significance analysis, and p-values < 0.05 was represented statistical significance. Student's *t*-test was used to compare the two means for differences, whereas Tukey's multiple comparisons were used for testing multiple comparisons.

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Author contributions

Guoqi Li and Xiaohong Cao designed the experiments. Guoqi Li, Xiaohong Cao and Qianhua Zeng performed the experiments. Guoqi Li, Xiaohong Cao and Shihua Wang analyzed the data. Guoqi Li, Elisabeth Tumukunde and Shihua Wang wrote the manuscript. The manuscript was discussed and revised by all the authors.

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Editors

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Reviewer #1 (Public Review):

In this manuscript, the authors investigated the roles of the target of rapamycin (TOR) pathway in various pathobiological processes of *Aspergillus flavus*. They found that rapamycin treatment affects the growth, sporulation, sclerotia, and aflatoxin synthesis of A.

flavus. The authors identified four immunophilin genes (FKBP1 -4), among which FKBP3 is involved in both rapamycin and FK506 resistances, with K19 residue being essential for succinylation. The authors identified a single Tor kinase and characterized its function. Subsequently, the authors analyzed a series of downstream effectors of the TOR pathway, including Sch9, TapA, SitA, Ppg1, and Spot7/Nem1, in terms of vegetative growth, sexual development, stress responses, and aflatoxin production.

While the authors provided a large amount of data regarding the genes involved in the TOR pathway, it is highly descriptive and mostly confirmative data, as numerous papers have already shown that the TOR pathway plays essential roles in a myriad of biological processes in multiple fungi. The authors seemed to perform a series of parallel studies in several genes involved in the TOR pathway in other fungi. However, their data are not properly interconnected to understand the TOR signaling pathway in this fungal pathogen. The authors frequently drew premature conclusions from basic phenotypic observations. For instance, based on their finding that *sch9* mutant showed high calcium stress sensitivity, they concluded that Sch9 is the element of the calcineurin-CrzA pathway. Furthermore, based on their finding that the *sch9* mutant show weak rapamycin sensitivity and increased Hog1 phosphorylation, they concluded that Sch9 is involved in TOR and HOG pathways. To make such conclusions, the authors should provide more detailed mechanistic data.

In the section "Tor kinase plays important roles in *A. flavus*", some parts of their data are confusing. The authors said they identified a single Tor kinase ortholog, which is orthologous to *S. cerevisiae* Tor2. And then, they said failed to obtain a null mutant, but constructed a single copy deletion strain Δ Tor1+/Tor2-. What does this mean? Does this mean *A. flavus* diploid strain? So is this heterozygous TOR/tor mutant? Otherwise, does the haploid *A. flavus* strain they used contain multiple copies of the TOR gene within its genome? What is the real name of *A. flavus* Tor kinase (Tor1 or Tor2?). "*tor1+/tor2-*" is the wrong genetic nomenclature. What is the identity of Δ tor1+/Tor2-? Please provide detailed information on how all these mutants were generated. A similar issue was found in the analysis of TapA, which is speculated to be essential (what is the Δ TapA1+/TapA2-?). I couldn't find any detailed information even in Materials and Methods. The authors should provide southern blot data to validate all their mutants.

How were the FRB domain deletion mutants constructed? If the FKBP12-rapamycin binding (FRB) domain is specifically deleted in the Tor kinase allele, should it be insensitive and resistant to rapamycin? However, the authors showed that the FRB domain deleted TOR allele was indeed non-functional.

In Figure 4C, the authors should monitor Hog1 phosphorylation patterns under stressed conditions, such as NaCl treatment, and provide quantitative measurements. Similar issues were found in the western blot analysis of Slt2 (Fig. 8D).

For all the deletion mutants generated in this study, the authors should generate complemented strains to validate their data.

- <https://doi.org/10.7554/eLife.89478.1.sa2>

Reviewer #2 (Public Review):

In this study, the authors identified the complex TOR, HOG, and CWI signaling networks-involved genes that relatively modulate the development, aflatoxin biosynthesis and pathogenicity of *A. flavus* by gene deletions combined with phenotypic observation.

They also analyzed the specific regulatory process and proposed that the TOR signaling pathway interacts with other signaling pathways (MAPK, CWI, calcineurin-CrzA pathway) to regulate the responses to various environmental stresses. Notably, they found that FKBP3 is

involved in sclerotia and aflatoxin biosynthesis and rapamycin resistance in *A. flavus*, and that the conserved site K19 of FKBP3 plays a key role in regulating the aflatoxin biosynthesis. In general, there is a heavy workload task carried in this study and the findings are interesting and important for understanding or controlling aflatoxin biosynthesis. However, findings have not been deeply explored and conclusions mostly are based on parallel phenotypic observations. In addition, there are some concerns that exist surrounding the conclusions.

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Reviewer #3 (Public Review):

The paper by Li et al. describes the role of the TOR pathway in *Aspergillus flavus*. The authors tested the effect of rapamycin in WT and different deletion strains. This paper is based on a lot of experiments and work but remains rather descriptive and confirms the results obtained in other fungi. It shows that the TOR pathway is involved in conidiation, aflatoxin production, pathogenicity, and hyphal growth. This is inferred from rapamycin treatment and TOR1/2 deletions. Rapamycin treatment also causes lipid accumulation in hyphae. The phenotypes are not surprising as they have been shown already for several fungi. In addition, one caveat is in my opinion that the strains grow very slowly and this could cause many downstream effects. Several kinases and phosphatases are involved in the TOR pathway. They were known from *S. cerevisiae* or filamentous fungi. The authors characterized them as well with knock-out approaches.

As for many results, I miss the re-complementation of the created mutants throughout the manuscript. This is standard praxis.

Fig. 1: cultures were grown for 48 h before measuring the transcript level. The authors show that *brlA*, *abaA*, and some sexual regulators are less expressed. In my opinion, this does not allow the conclusion that there is a direct control through rapamycin. Since the colonies grow very slowly in the presence of rapamycin, the authors should add rapamycin and follow gene expression after 15, 30, 60, 90 min. The figure legend needs to be more detailed. Which type of cultures were used, liquid, solid medium? Etc.

Why in chapter one Fig. 9 is already cited? Those data should then be included in Fig. 1 for the general phenotype.

The authors wrote that radial growth and conidiation were gradually reduced with increasing rapamycin concentrations. This is not true. There is no gradient! However, it should be tested if there is a gradient if lower concentrations are used. The current data imply that there is a threshold concentration, so either there is 100 % growth or a reduction to 25 %. This looks strange.

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